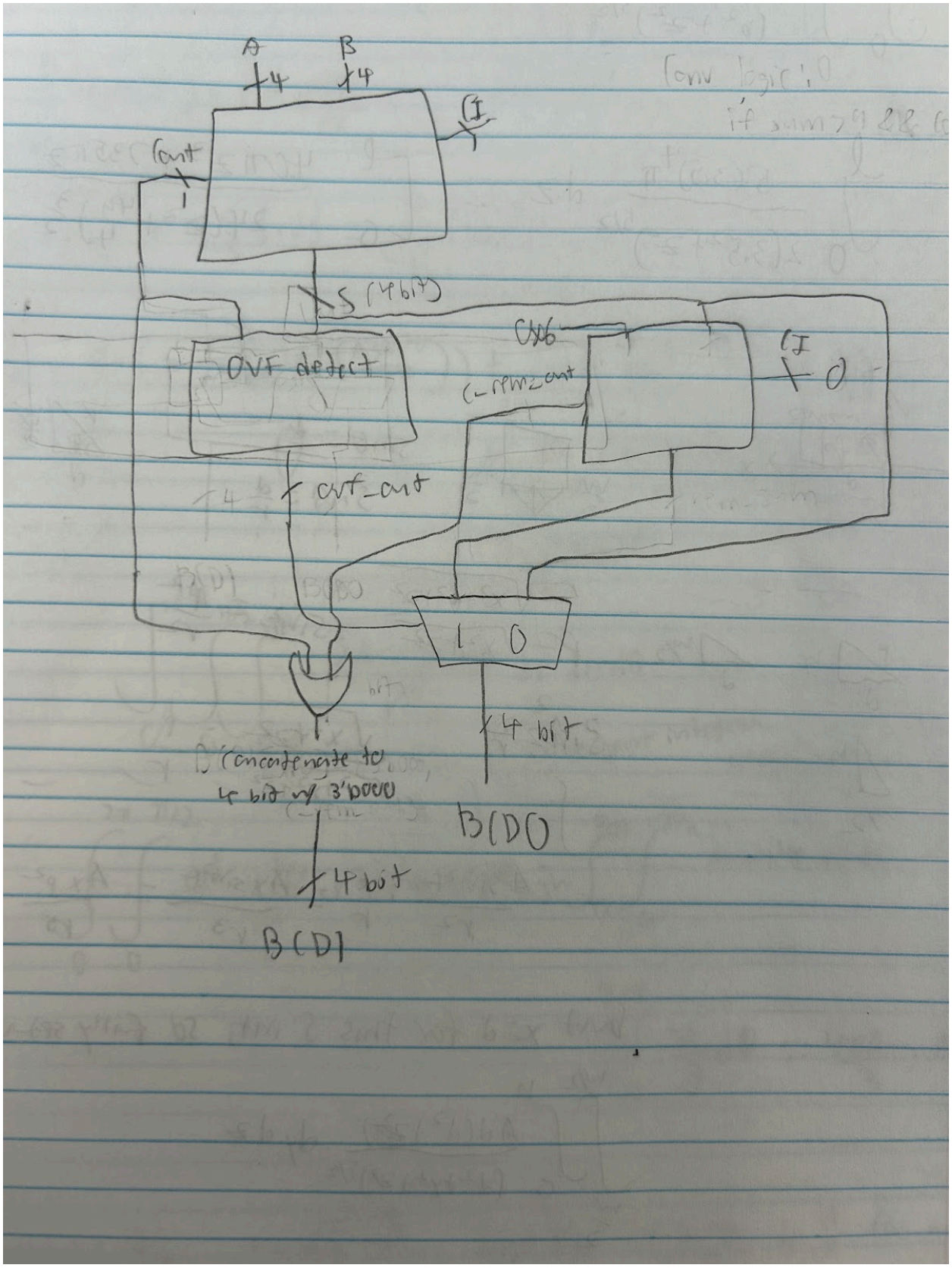
**Dylan Thornburg CSEN 127 Lab 2 Prelab**

Verilog:

```
module bcdadd (
	input [3:0] A,
	input [3:0] B,
	input Ci,
	output [3:0] bcd0,
	output [3:0] bcd1
);
	wire [3:0] S;
	wire Cout;
	wire [3:0]addr2out;
	wire C_rem_out;
	adder4 one (A,B,Ci,Cout,S);
	adder4 two (4'b0110,S,4'b0000,C_rem_out, addr2out);
	wire ovf_out;
	assign ovf_out = Cout || (S > 4'b1001);
	assign bcd0 = ovf_out ? addr2out : S;
	assign bcd1 = {3'b000, (C_rem_out)}; //  C_rem_out | Cout to make it work for a | b >9
endmodule
```

Block Diagram:


A
B
4
4
CI
Cont
1
S (4bit)
OVF detect
CI
0
ovf_out
1
0
4 bit
Concatenate to
4 bit w/ 3'b000
B(D0)
4 bit
B(D1)

**I misread the initial prompt. An or gate is not necessary and it should just be C_rem_out going to BCD1. Having it “ored” with Cout allows inputs >9 to work, but these are undefined states, so it’s ultimately better prioritize less gates.**